



LIGHT BEAMS AND JOISTS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
mm (inch)	cm ⁴	in ⁴	cm ⁴	in ⁴	cm	in	cm	in	cm ³	in ³	cm ³	in ³
76 x 51 (3 x 2)	83	2.0	11.1	0.27	3.12	1.23	1.14	0.45	21.67	1.33	4.37	0.27
102 x 64 (4 x 2½)	218	5.2	25.3	0.61	4.21	1.66	1.43	0.57	42.84	2.62	7.97	0.49
102 x 76 (4 x 3)	325	7.8	55.4	1.33	4.14	1.63	1.70	0.67	63.7	3.89	14.4	0.88
127 x 76 (5 x 3)	476	11.5	50.2	1.21	5.29	2.08	1.72	0.68	74.9	4.58	13.17	0.81
152 x 76 (6 x 3)	874	21.5	60.8	1.46	6.20	2.44	1.63	0.64	114.7	7.00	15.9	0.97
152 x 89 (6 x 3½)	881	21.7	86.0	2.07	6.36	2.51	1.99	0.78	115.6	7.07	19.34	1.18
152 x 127 (6 x 5)	1,819	43.7	378.7	9.10	6.20	2.44	2.82	1.11	239.1	14.56	59.6	3.64
178 x 102 (7 X 4)	1,519	36.6	139.2	3.36	7.44	2.93	2.25	0.89	170.9	10.45	27.41	1.68
178 x 102 (7 x 4)	1,644	39.5	140.3	3.37	7.34	2.89	2.13	0.84	185.0	11.29	27.7	1.69
203 x 102 (8 x 4)	2,294	55.1	162.6	3.91	8.43	3.32	2.25	0.88	225.8	13.78	32.02	1.95
203 x 102 (8 x 4)	2,314	55.6	146.1	3.51	8.23	3.24	2.06	0.81	227.9	13.91	28.7	1.75
203 x 152 (8 x 6)	4,790	115.1	813.3	19.54	8.48	3.34	3.51	1.38	471.3	28.76	106.7	6.51
228 x 102 (9 x 4)	3,375	81.1	172.7	4.15	9.19	3.62	2.08	0.82	295.5	18.03	33.9	2.07
254 x 127 (10 x 5)	6,085	146.2	405.0	9.73	10.31	4.06	2.67	1.05	479.3	29.25	63.7	3.89
254 x 152 (10 x 6)	8,524	204.8	905.6	21.76	10.59	4.17	3.45	1.36	671.2	40.96	118.8	7.25
254 x 203 (10 x 8)	12,016	288.7	2,278.3	54.74	10.72	4.22	4.67	1.84	946.2	57.74	224.3	13.69
305 x 127 (12 x 5)	9,202	221.1	403.3	9.69	12.29	4.84	2.56	1.01	603.7	36.84	63.6	3.88
305 x 152 (12 x 6)	13,181	316.7	920.6	22.12	12.55	4.94	3.30	1.30	865.1	52.79	120.8	7.37